

PROJECT REPORT

Course: Machine Learning for Robotics

Section: BSC-8M

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Metro Interstate Traffic Volume Prediction Project

1. Introduction and Problem Statement

Traffic congestion is a major issue in urban areas, leading to increased travel time, fuel consumption, and environmental pollution. Accurate prediction and analysis of traffic volume can help in better traffic management, route planning, and intelligent transportation systems.

The objective of this project is to analyze the Metro Interstate Traffic dataset and apply multiple machine learning techniques to:

- Classify traffic levels (low, medium, high)
- Predict traffic volume using regression
- Identify hidden patterns using clustering
- Simulate decision-making using reinforcement learning

This project also evaluates and compares different machine learning algorithms to determine their effectiveness and suitability for real-world and robotics applications.

2. Dataset Description and Preprocessing Steps

Dataset Description

The dataset used is the *Metro Interstate Traffic Volume Dataset*, which contains hourly traffic data along with weather and temporal features.

Key features include:

- Date and time information
- Traffic volume (target variable)
- Weather conditions (temperature, rain, snow, clouds)
- Holiday information

Preprocessing Steps

Several preprocessing steps were applied to ensure data quality and improve model performance:

1. Handling Missing Values

- Missing values were filled using forward fill method.

2. Removing Duplicates

- Duplicate rows were removed to avoid bias.

3. Datetime Conversion

- The `date_time` column was converted into datetime format.

4. Feature Engineering

- Extracted features such as year, month, day, hour, and day of week.
- Created additional features:
 - `is_rush_hour`
 - `is_weekend`

5. Encoding Categorical Variables

- Label encoding was applied to:
 - `weather_description`
 - `holiday`
 - `weather_main`

6. Weather Severity Mapping

- Weather conditions were mapped into severity levels.

7. Normalization

- Min-Max scaling was applied to numerical features such as temperature, rain, snow, and cloud coverage.

8. Train-Test Split

- The dataset was split into training (80%) and testing (20%) sets

3. Model Implementation Details

3.1 Classification Models

Traffic volume was converted into three categories (low, medium, high) using quantile-based binning.

The following classification models were implemented:

- **K-Nearest Neighbors (KNN)**
 - Classifies based on nearest neighbors using distance metrics.
- **Support Vector Machine (SVM)**
 - Finds an optimal hyperplane to separate classes.
- **Decision Tree**
 - Uses a tree-like structure for decision making.
- **Bagging (Ensemble Learning)**
 - Combines multiple decision trees trained on random subsets to improve performance and reduce overfitting.

3.2 Regression Model

- **Linear Regression**
 - Predicts continuous traffic volume using a linear relationship between features and target.

3.3 Clustering Model

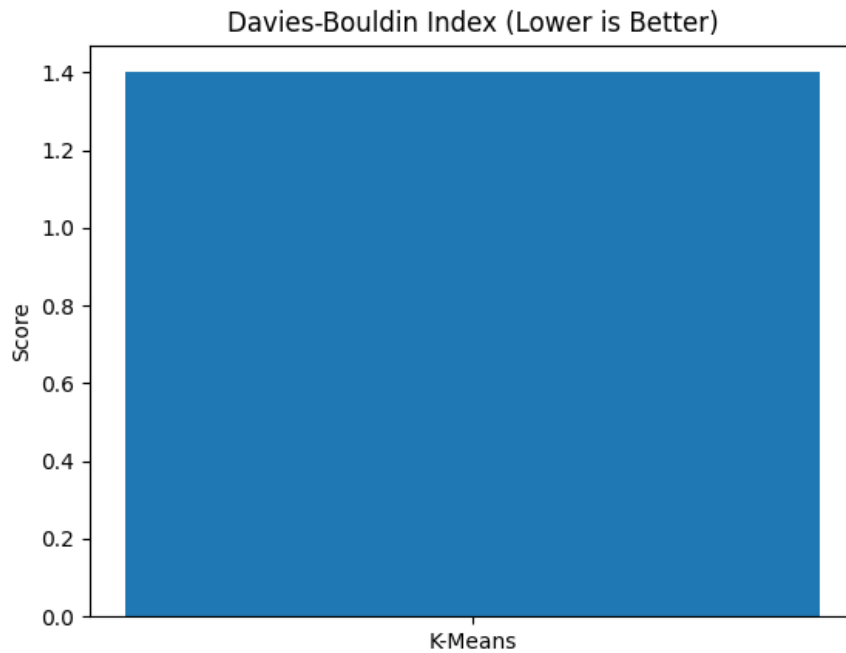
- **K-Means Clustering**
 - Groups data into clusters based on similarity.
 - Used to identify hidden traffic patterns.

3.4 Reinforcement Learning

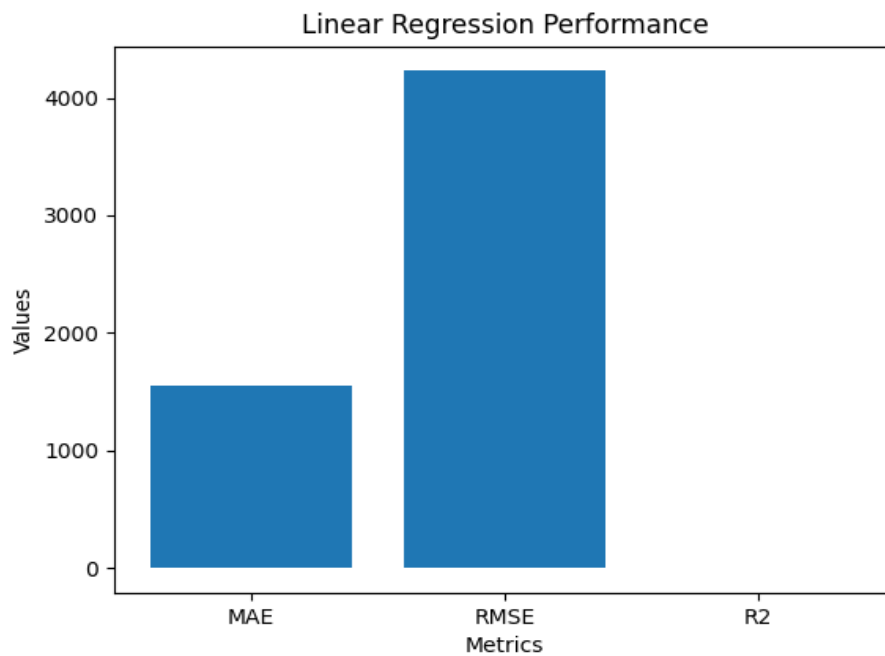
- **Q-Learning**
 - A model-free reinforcement learning algorithm.
 - Simulates traffic signal decision-making based on traffic conditions.
 - Uses a Q-table to learn optimal actions.

4. Comparative Analysis and Conclusion

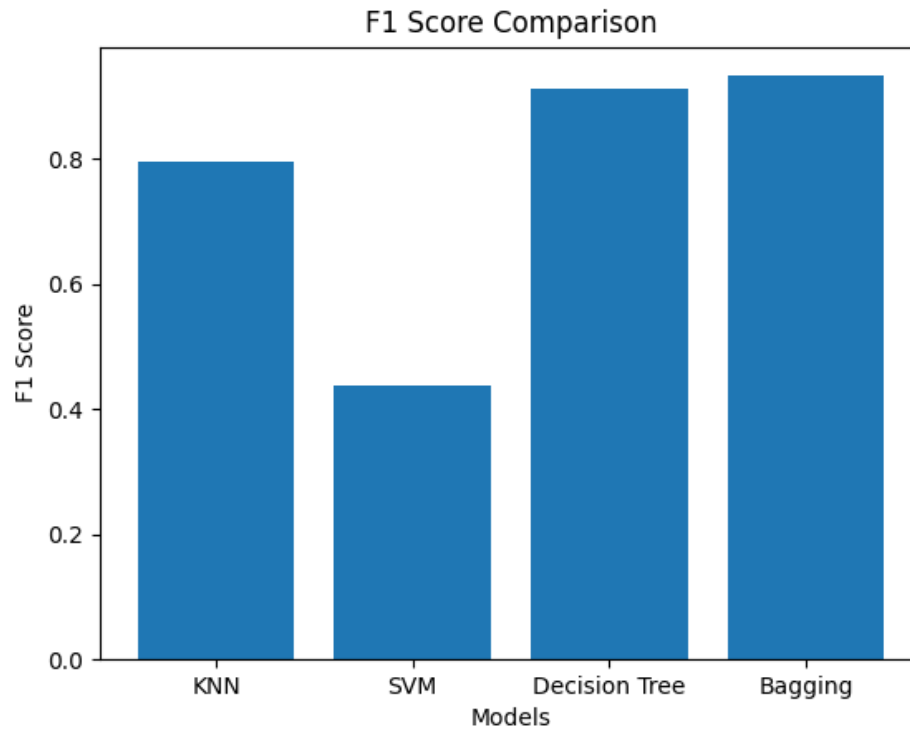
4.1 Davies-Bouldin Index



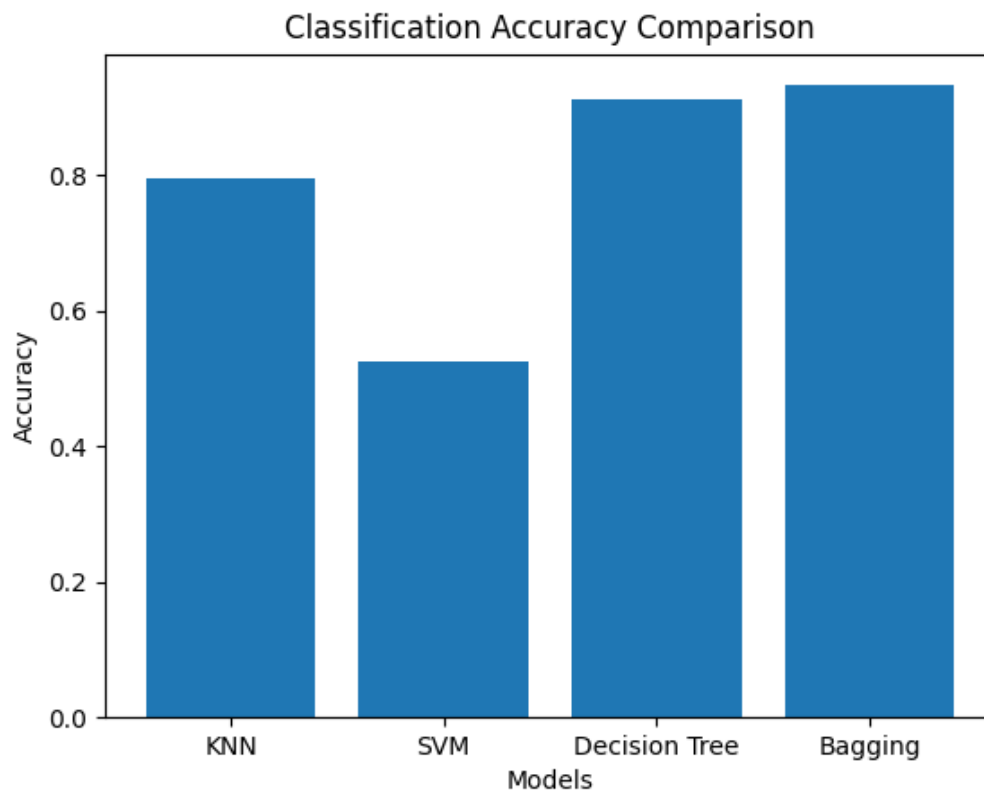
4.2 Linear Regression Performance



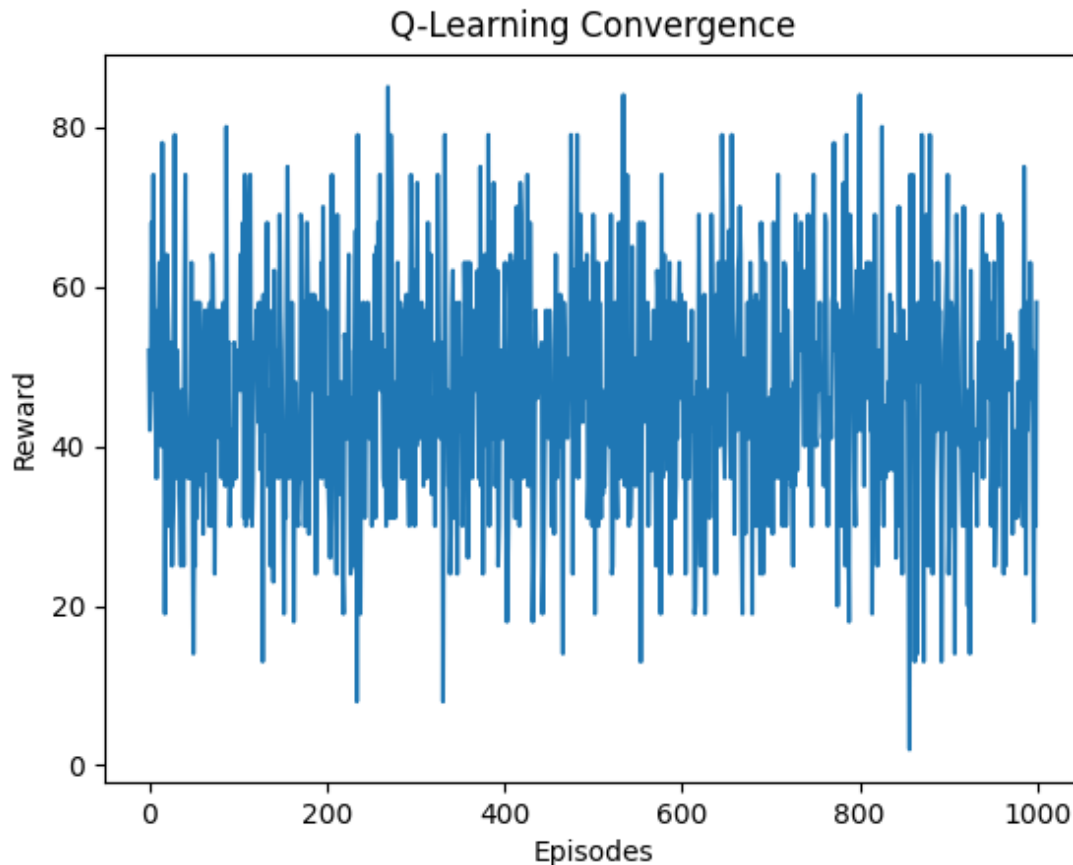
4.3 F1 Score Comparison



4.4 Classification Accuracy Comparison



4.5 Q-Learning Convergence



5. Comparative Analysis and Conclusion

5.1 Performance Comparison

- **Classification Models**
 - Bagging showed improved accuracy compared to individual models due to reduced variance.
 - Decision Tree performed well but was prone to overfitting.
 - SVM provided stable results but required more computation.
 - KNN performance depended heavily on parameter selection and scaling.
- **Regression Model**
 - Linear Regression provided a baseline model with reasonable performance.
 - Performance evaluated using MAE, RMSE, and R^2 .

- **Clustering Model**
 - K-Means clustering performance was evaluated using Davies-Bouldin Index.
 - Lower values indicated better cluster separation.
- **Reinforcement Learning**
 - Q-Learning showed improvement in cumulative reward over episodes.
 - Convergence indicated learning stability.

5.2 Computational Complexity

- KNN: High prediction cost
- SVM: High training cost
- Decision Tree: Fast and efficient
- Bagging: Moderate (multiple models)
- Linear Regression: Very fast
- K-Means: Moderate
- Q-Learning: Depends on number of episodes

5.3 Suitability for Robotics Applications

- Decision Tree and Bagging: Suitable for fast decision-making
- Linear Regression: Useful for prediction tasks
- K-Means: Useful for environment grouping
- Q-Learning: Highly suitable for adaptive and autonomous systems

Conclusion

This project demonstrates the application of multiple machine learning techniques on traffic data. Ensemble methods such as Bagging improved classification performance, while regression and clustering provided additional insights into traffic patterns. Reinforcement learning showed potential for adaptive decision-making, making it highly relevant for robotics and intelligent transportation systems.

Overall, different models serve different purposes, and selecting the appropriate model depends on the problem requirements, computational constraints, and application domain.